Indian Institute of Technology Jodhpur

Concept note and Curriculum of

M. Tech.

in

Microwave and Photonic Technologies



Background:

The rapid evolution of modern communication, defence & aerospace, sensing, and imaging technologies has significantly driven progress in both microwave and photonic technologies, underlining their individual importance across a wide range of applications. There is a growing global need for professionals in microwave and photonic technology domain including the areas like 5G and 6G wireless networks, satellite communications, quantum photonics, RADAR, LiDAR, and terahertz imaging. The department of **Electronics Engineering** at IIT Jodhpur is proposing a comprehensive **M.Tech.** program in **Microwaves and Photonic Technologies** to address national and global challenges and opportunities in the above-mentioned areas. This proposed program stands on fundamental concepts including electromagnetic theory, microwave engineering, optical systems, and prioritizing cutting-edge research, practical experiential learning, and robust collaboration between industry and academia.

Program justification and relevance

- **Research and innovation:** Emerging fields like MMIC and RFIC design, advanced metasurfaces and antennas, RF front-end for communication systems and RADARs, photonic circuits and devices, optical communication, and photonic sensors offer significant opportunities for impactful research and development.
- **Industry demand:** The telecommunication, defence, aerospace, and semiconductor industries need skilled engineers who can design and develop microwave and photonic components and systems.
- **Strategic value for emerging India:** Technologies based on microwave and photonics are essential for national security, space exploration, high speed wireless links, and smart infrastructure development.

Program Title	Duration	Degree Name	Intake
Microwave and Photonic Technologies	02 Years	M. Tech.	Total: 15 (With MoE fellowship: 08) (Without fellowship + Self-sponsored: 07)

Eligibility criteria:

i. With MoE fellowship

Qualifying Degree	GATE Requirement	Marks/CPI requirement
B. Tech., B.E. or B.S/ 04 years B.Sc. (Engineering) in Electronics and Communication Engineering/Electrical Engineering/Electronics and Telecommunication Engineering/Electrical and Electronics Engineering /Electronics engineering or M.Sc. in Electronics or equivalent	Valid GATE score in EC	Minimum of 60% or 6.0 (on a 10-point scale) for GE/EWS/OBC Minimum of 55% or 5.5 (on a 10-point scale) for SC/ST/PWD
B. Tech., B.E. or B.S/ 04 years B.Sc. (Engineering) in Electronics and Communication Engineering/Electrical Engineering/Electronics and Telecommunication Engineering/Electrical and Electronics Engineering /Electronics engineering or equivalent from any of the CFTIs with CPI or CGPA of 8/10 or above	The GATE score requirement is exempted	

ii. Without fellowship

Qualifying Degree	GATE Requirement	Marks/CPI requirement
B. Tech., B.E. or B.S/ 04 years B.Sc. (Engineering) in Electronics and Communication Engineering/Electrical Engineering/Electronics and Telecommunication Engineering/Electrical and Electronics Engineering /Electronics engineering or M.Sc. in Electronics or equivalent	Valid GATE score in EC	Minimum of 60% or 6.0 (on a 10-point scale) for GE/EWS/OBC. Minimum of 55% or 5.5 (on a 10-point scale) for SC/ST/PWD.

The selection will be done based on the performance in the written test/interview conducted by the department.

iii. Self-sponsored

Qualifying Degree	GATE Requirement	Marks/CPI requirement
B. Tech., B.E. or B.S/ 04 years B.Sc. (Engineering) in Electronics and Communication Engineering/Electrical Engineering/Electronics and Telecommunication Engineering/Electrical and Electronics Engineering /Electronics engineering or M.Sc. in Electronics or equivalent	Not required	Minimum of 60% or 6.0 (on a 10-point scale) for GE/EWS/OBC. Minimum of 55% or 5.5 (on a 10-point scale) for SC/ST/PWD.

The selection will be done based on the performance in the written test/interview conducted by the department.

Objectives of the Program:

The primary objective of this program is to cultivate advanced technical learning and R&D oriented skills in microwave, millimeter-wave, photonic and optical systems. This program will prepare students to pursue careers in the industries (aerospace, defence and emerging communication systems), research organizations and academia. Following are the key objectives of the proposed M.Tech. program.

1. To equip students with strong theoretical foundations in the subject matters.
2. Hands-on training in the design, analyses, testing of the microwave and photonic systems using modern tools/software.
3. To encourage innovative problem-solving abilities in the emerging field of RF-microwave and photonics.
4. To train the students on modern CAD tools, cutting edge equipment used in microwave and photonics.

Graduate Attributes

- In depth theoretical and practical understanding of EM wave propagation, RF-microwave active and passive circuits, photonic devices.
- Ability to design, analyze, and characterize circuits and systems at microwave, millimetre-wave and optical frequencies.
- Proficiency in industry standard simulators like Ansys HFSS, CST, Agilent ADS, MATLAB.
- Ability to handle academic and industrial projects.

Learning Outcomes

Graduates will gain knowledge in the field of

- In-depth knowledge of RF-microwave and millimeter-wave components, circuits and systems, RF front-end for modern communication systems, antenna arrays, advanced metasurfaces, guiding structures (both microwave and optical), photonic circuits, imaging systems.
- Hands-on experience with high-end microwave/mm-wave and optical equipment.

- Knowledge of various CAD tools like CST Microwave Studio, Ansys Electronics Desktop (HFSS), COMSOL, Keysight ADS, CADENCE, and MATLAB.
- Proficiency in designing of RFICs, MMIC, and photonic circuits.
- Knowledge in Printed Circuit Board (PCB) layout designing and prototyping, cleanroom fabrication techniques.

Program Structure

S.No.	*Category*	*Course category title*	*Total Courses*	*Total Credits*
1	C	*COMPULSORY*	6	14
2	E	*ELECTIVES*	4	12
3	NG	*Non-Graded*	2	0
4	T	*Thesis*	2	32
5	T	*Thesis during summer break*	1	8
Total				**66**

The thesis work must be aligned with the objectives of the enrolled program as stated in the concept note.

Semester-wise Distribution of Credits

Cat.	*Course Number, Course Title*		*L-T-P*	*Credits*		*Cat.*	*Course Number, Course Title*		*L-T-P*	*Credits*
Semester-I						**Semester-II**				
C	ECL7XX0	Advanced Electromagnetics	3-0-0	3		C	ECL7XX0	Microwave and Optical Measurements	3-0-0	3
C	ECL7XX0	Microwave Circuits	3-0-0	3		C	ECL7XX0	Microwave and Optical Measurements Lab.	0-0-2	1
C	ECL7XX0	Fiber Optic Devices and Components	3-0-0	3		E	ECL7XX0	*Elective*	3-0-0	3
C	ECL7XX0	Microwave Circuits Lab.	0-0-2	1		E	ECL7XX0	*Elective*	3-0-0	3
E	ECL7XX0	Elective	3-0-0	3		E	ECL7XX0	*Elective*	3-0-0	3
NG	LAN7XX0	Technical communication	1-0-0	S/U		NG	OAL7XX0	Innovation and IP management	1-0-0	S/U
			Total	**13**					*Total*	**13**
						Summer term				
						ECT7XXX		Thesis		8
									Total	**8**
III Semester						**IV Semester**				
T	ECT7XXX	Thesis		16		T	ECT7XXX	Thesis		16
			Total	**16**					*Total*	**16**
Total credit										**66**

List of Electives

S.No.	Course Number	*Course Title*	L-T-P	Credit
1		*Antennas and Wave Propagation*	3-0-0	3
2		*RFIC and MMIC*	3-0-0	3
3		*Millimeter-Wave and THz Technologies*	3-0-0	3
4		*RF system design for Communications*	3-0-0	3
5		*RADAR Engineering*	3-0-0	3
6		*Optical Signal Processing*	3-0-0	3
7		*Computational Electromagnetics*	3-0-0	3
8		*Photonic Devices and Circuits*	3-0-0	3
9		*Optical Communication Systems*	3-0-0	3
10		*Optical Networks*	3-0-0	3
11		*IR Detectors and Imaging*	3-0-0	3
12		*Machine Leaning*	*3-0-0*	3

Detailed Course Contents

Compulsory courses

Title	Advanced Electromagnetics	Number	ECL7XX
Department	Electronics Engineering	L-T-P [C]	3–0–0 [3]
Offered for	B.Tech. (IV Year), M.Tech., and PhD	Type	C
Prerequisite	Engineering Electromagnetics		

Objectives

1. To gain the knowledge of advanced Electromagnetics.
2. To familiarize the students with radiation, wave propagation in unbounded medium, inside waveguides, and transmissions line.

Learning Outcomes

1. Ability to solve EM wave equations, design of waveguides.
2. Understanding the transmission lines.
3. Ability to compute modes and analyze the propagation behavior in dielectric waveguides.

Contents

Introduction to EM wave [10 lectures**]**
Concept of E, D, B, and H, Boundary conditions, polarization of dielectric and displacement vector, Maxwell's equations, homogeneous and inhomogeneous wave equations and its solutions, EM theorems

Interaction of EM wave in different medium [8 lectures**]**
Types of Medium, EM wave propagation in unbounded medium, EM scattering, Geometric theory of diffraction, Green's function and integral transform techniques

Waveguides [12 lectures**]**
Modes inside rectangular and circular waveguides, cut-off frequency, wave impedance, loss inside waveguide, waveguide resonators, losses in a resonator, resonator Q-factor, partially filled waveguide, ridge waveguide

Optical planar waveguides [9 lectures]
Reflection and Refraction, Fresnels' law, Ray propagation, Goos-Haenchen Shift, Asymmetric dielectric slab waveguide, Effective width, group velocity, Wave propagation, TE and TM modes, dispersion relations, cut-off, power flow, weakly guiding approximation

Textbooks:
1. D. J. Griffiths, "Introduction to Electrodynamics," 3rd edition, Prentice-Hall (1995).
2. R. E. Collin: Field theory of Guided Waves, Wiley IEEE press.
3. R. F. Harrington: Time Harmonic Electromagnetic Fields, McGraw-Hill, 1961
4. Yariv and Yeh, Photonics: Optical Electronics in Modern Communication, 2006. Oxford Series in Electrical and Computer Engineering

Self-learning Materials
1.http://ocw.mit.edu/OcwWeb/Physics/8-02Spring-2007.Electromagnetic theory NPTEL (IIT Kanpur), https://onlinecourses.nptel.ac.in/noc18_ee04/preview

Preparatory Course Materials
1. MNO Sadiku, Elements of Electromagnetics, Oxford Univ. Press.

Title	**Microwave Circuits**	**Number**	**ECL7XX**
Department	Electronics Engineering	L-T-P [C]	3–0–0 [3]
Offered for	B.Tech. (IV Year), M.Tech., and PhD	Type	C
Prerequisite	Engineering Electromagnetics, Analog circuits		

Objectives

1. To introduce the fundamentals of transmission lines and microwave/mm-wave circuit analysis.
2. To introduce the students with the active circuit designing at microwave and millimeter-wave frequencies.

Learning Outcomes

1. Understanding the design of integrated circuits at high frequencies.
2. Practical implementation of various circuits required to build a transceiver system.
3. Learning about the design of various active components like mixers, LNAs, power amplifiers at RF and microwave frequencies.

Contents

Fundamentals of transmission line and matching networks [6 lectures**]**
Overview of transmission lines, coaxial cables, power flow and losses, transients, microstrip transmission line, co-planar waveguide (CPW), coaxial line, losses and cut off frequencies, S-parameters, signal flow graphs, impedance matching techniques, microwave network analysis.

Three and four port microwave networks [6 lectures**]**
Wilkinson power divider, hybrid coupler, directional coupler, power combiners, circulators, attenuators, Magic-Tees.

Amplifier design [8 lectures**]**
Concepts of nonlinearity, IM products, third order intercepts, model of MOS transistors and BJT, HEMT at high frequencies, stability analyses, design of biasing and stability networks, transmission line realization, design of maximum gain amplifier, wideband amplifiers, constant gain amplifier, low noise amplifier (LNA), power amplifiers (PA), combining networks.

Oscillators and Voltage Controlled Oscillators (VCO) [6 lectures**]**
Basic topologies of oscillators, phase noise, design of oscillator, VCO, Noise-power trade-off, quadrature, and single-sideband generators.

Mixers [7 lectures**]**
Mixer Noise figure, diode based mixers, FET mixer, Gilbert cell mixer, port to port feed-through, single-balanced and double balanced mixers.

Introduction to Phase-Locked Loops (PLL) [6 lectures**]**
Basic architecture, phase detector, loop stability, tracking, Type I and Type II PLL's, PLL Non-idealities, phase noise, jitter.

Textbooks

1. T. H. Lee, the Design of CMOS Radio-Frequency Integrated Circuits. Cambridge, UK: Cambridge University Press, 1997.
2. B. Razavi, RF Microelectronics, 2nd Edition, Pearson Education, 2014.
3. G. Gonzalez, Microwave Transistor Amplifiers: Analysis and Design, 2nd ed., Pearson,Aug-1996.

Self-learning Materials

1. RF Integrated Circuits, NPTEL, Dr. S. Chatterjee, IIT Delhi, http://nptel.ac.in/courses/117102012/#
2. High Speed Communication Circuits, MIT, https://ocw.mit.edu/courses/electrical-engineering-and-computer-science/6-776-high-speed-communication-circuits-spring-2005/lecture-notes/

Preparatory Course Materials

1. D. M. Pozar, Microwave Engineering, 4th edition, New York, NY, USA: Wiley, 2011.

Title	**Microwave Circuits Lab**	**Number**	**ECL7XX**
Department	Electronics Engineering	L-T-P [C]	0–0–2 [1]
Offered for	B.Tech. (IV Year), M.Tech., and PhD	Type	C
Prerequisite	Engineering Electromagnetics, Analog circuits		

Objectives

1. To introduce the students with the active circuit designing using CAD tools, 3-D simulators, circuit simulators at microwave and millimeter-wave frequencies.

Learning Outcomes

1. Working with high frequency circuit design CAD tools.
2. Ability to design matching networks, microwave circuit components like low noise amplifier, Power amplifier, scillators, mixers.

Contents

Introduction to HFSS, CST, ADS. Modelling and simulations of discrete circuit components, design of matching networks, maximum and specific gain amplifier design, LNA design, PA design.

Textbooks

1. B. Razavi, RF Microelectronics, 2nd Edition, Pearson Education, 2014.
2. G. Gonzalez, Microwave Transistor Amplifiers: Analysis and Design, 2nd edition, Pearson, Aug-1996.

Self-learning Materials

1. RF Integrated Circuits, NPTEL, Dr. S. Chatterjee, IIT Delhi, http://nptel.ac.in/courses/117102012/#
2. High Speed Communication Circuits, MIT, https://ocw.mit.edu/courses/electrical-engineering-and-computer-science/6-776-high-speed-communication-circuits-spring-2005/lecture-notes/

Preparatory Course Materials

1. D. M. Pozar, Microwave Engineering, 4th edition, New York, NY, USA: Wiley, 2011.

Title	**Fiber Optic Devices and Components**	**Number**	**ECL7XX**
Department	Electronics Engineering	L-T-P [C]	3–0–0 [3]
Offered for	B.Tech, M.Sc. Physics, M.Tech. and PhD	Type	C
Prerequisite	Engineering Electromagnetics		
Objectives			

1. Provide the fundamental principles of fiber waveguides and various devices and components based on optical fiber.
2. Enable the appreciation of the developments in optical fiber technology.

Learning Outcomes
The students are expected to have the ability to implement a fiber optic link using various devices, components, and various modules for different requirements.

Contents
Optical fibers [20 Lectures**]**
Vector Nature of Light, Optical Fiber – Single and Multimode, Step index and Graded Index Fibers, Ray and Wave picture of propagation, Cut-off, mode field diameter, group delay, Gaussian field approximation. Propagation characteristics of few modes, multicore and photonic crystal fibers. Fiber Fabrication Methods.
Transmission Characteristics: Attenuation and dispersion in Optical Fibers, Material, Waveguide, Polarization, Intermodal and Intramodal Dispersion, Group velocity Dispersion, Dispersion modified single mode fibers
Link components [16 Lectures**]**
Fiber joints, connectors, Splicing operation, OTDR, Design of splitter, couplers, Fiber Bragg Gratings, Isolators, Circulators, Multimode Interference devices, mode field convertors, wavelength MUX/DEMUX, fiber to chip coupling, Fiber Amplifiers: EDFA, Raman and Brillouin, SOA.
System planning [3 Lectures**]**
Choice of components for optical fiber communication, Link Design, Impact of nonlinear effects on communication systems.

Textbooks
1. J. M. Senior, Optical Fiber Communications: Principles and Practice, Prentice Hall, 3rd Edition, 2010.
2. Raman Kashyap, Fiber Bragg Gratings, Academic Press, 2009.
3. G. Agrawal, Fiber optic Communication Systems, John Wiley and sons, 4th Edition 2010.

Self-learning Material
1. NPTEL Lectures on Advanced Optical Communication by Prof. R.K. Shevgaonkar, IIT Bombay(https://nptel.ac.in/courses/117101002/)
2. The RP Photonics Encyclopedia (https://www.rp-photonics.com/encyclopedia.html)

Preparatory Course Material:
1. Concepts from "M.O. Sadiku and S.V. Kulkarni, Principles of Electromagnetics", Oxford University Press, 6th edition, 2015.

Title	**Microwave and Optical Measurements**	**Number**	**ECL7XX**
Department	Electronics Engineering	L-T-P [C]	3–0–0 [3]
Offered for	B.Tech. (IV Year), M.Tech., and PhD	Type	C
Prerequisite	Microwave Engineering		

Objectives
1. To introduce the students with the measurement equipment used in microwave to optical domain.s

2. To introduce the students with the measurement techniques at microwave and optical frequencies.

Learning Outcomes
1. Working and calibration of high-end measurement instruments used for the measurements at microwave and optical frequencies.
2. Understanding the measurements of integrated circuits, optical modulators, detectors.

Contents
Overviews [5 lectures**]**
Review of S-parameters, X-parameters, effects of non-linarites, noise floor, dynamic range, spurious free dynamic range.
Basics of calibration techniques [2 lectures**]**
Overview of signal flow graph, basics of reflectometer, error models, scalar calibration, vector calibration, TRL calibration.
Scalar analyzer [5 lectures**]**
Architecture of scalar analyzer, working principle, Reflection bridge and calibration
Vector analyzer [8 lectures**]**
Architecture of VNA, source, signal separation, receiver module, reference plane and de-embedding, Vector measurement system, twelve term error model, signal detection.
Six ports [2 lectures**]**
General six port network calibration, multi state reflectometer
Noise figure [4 lectures**]**
Noise figure measurement, Y-factor method, cold source method, noise figure meter calibration, noise measurement by spectrum analyzer
Optical measurement techniques [13 lectures**]**
Optical measurement techniques for different physical parameters using Interferometric methods. Passive device testing. Fiber operation techniques such as OTDR and Splicing. Free-space optical beam characterization, propagation, parameter measurements, and data processing. Characterization of Laser Sources, photodetectors, and amplifiers. Implementation of techniques such as WDM and SDM. Optical Modulation Techniques. Holography. Quantum nature of light.

Textbooks
1. G.H. Bryant, Principles of Microwave Measurements, IEE Electrical measurement series
2. Niknejad Ali Electromagnetics for High-Speed Analog and Digital Communication Circuits. Cambridge University Press, 2007
3. Michael Bass - Handbook of Optics, Vol. 2, Devices, Measurements, and Properties-McGraw-Hill Professional (1994).
4. Ajoy K. Ghatak, K. Thyagarajan, An Introduction to Fiber Optics, Cambridge University Press, 1998.

Preparatory Course Materials
1. D. M. Pozar, Microwave Engineering, 4th edition, New York, NY, USA: Wiley, 2011.

Title	**Microwave and Optical Measurements Lab**	**Number**	**ECL7XX**
Department	Electronics Engineering	L-T-P [C]	0–0–2 [1]
Offered for	B.Tech. (IV Year), M.Tech., and PhD	Type	C
Prerequisite	Microwave Engineering		

Objectives
1. To introduce the students with the high-end measurement instruments.

2. Measurement and characterization of microwave, millimeter wave and optical circuit components.

Learning Outcomes
1. Handling the high frequency measurement instruments like VNA, power meter, waveform generators, optical detectors.
2. Measurements associated with microwave amplifiers, mixers, RADAR, optical modulators.

Contents
Calibration of VNA (SOLT and TRL), familiarization with spectrum analyzer and RF generators, optical spectrum analyzers, power meters, ultra-narrow pulse measurements, coherent RADAR measurement, optical modulators, optical detector circuits, antenna measurement, measurements of multiport microwave and optical passive devices, measurement of LNA, mixer characterization, microwave PA measurement.

Textbooks
1. G.H. Bryant, Principles of Microwave Measurements, IEE Electrical measurement series
2. Michael Bass - Handbook of Optics, Vol. 2, Devices, Measurements, and Properties-McGraw-Hill Professional (1994).
3. Ajoy K. Ghatak, K. Thyagarajan, An Introduction to Fiber Optics, Cambridge University Press, 1998.

Preparatory Course Materials
1. D. M. Pozar, Microwave Engineering, 4th edition, New York, NY, USA: Wiley, 2011.

Electives

Title	Antennas and Wave Propagation	Number	ECL7XX
Department	Electronics Engineering	L-T-P [C]	3–0–0 [3]
Offered for	B.Tech. IV Year, M. Tech., PhD	Type	E
Prerequisite	Engineering Electromagnetics		

Objectives

1. Introduce the various concepts of antenna design.
2. familiarize the students with the application of different type of antennas

Learning Outcomes

1. Design low profile antennas and high gain antenna arrays for practical applications
2. Utilize simulation software for designing Antenna systems and test RF and microwave antennas

Contents

Introduction to Antennas [4 lectures]
Radiation mechanism, near and far field, Review of Antenna Parameters; Friis Transmission formula, Radar Range equation

Radiation integrals and auxiliary potential function [4 lectures]
Vector potential for electric current source and magnetic current source, inhomogeneous vector potential wave equation, duality theorem, reciprocity theorem

Radio wave propagation [6 lectures]
Ground wave propagation, Sky wave and space wave propagation, atmospheric effects, ground effects, plasma effects, attenuation, scattering, diffraction, link budget, link margin.

Linear wire and loop antenna [7 lectures]
Infinitesimal dipole, small dipole, half-wavelength dipole, loop antenna

Antenna arrays [5 lectures]
Analysis of Two element array, N-element linear array (uniform and nonuniform amplitude excitation: Binomial, Dolph-Tschebyscheff, Taylor distribution), planar array, Phased Array Antennas

Aperture antenna [4 lectures]
Field equivalence principle, Fourier series technique, Radiation equation, Rectangular aperture, circular aperture, Babinet's principle, E-Plane Horn Antenna, H-Plane Horn Antenna, Pyramidal Horn antenna

Microstrip and dielectric resonator antennas [4 lectures]
Rectangular patch antenna, feeding methods, transmission line model, cavity model, Metallic vs Dielectric Antennas, Dielectric Resonator Antennas.

MIMO antennas [2 lectures]
Single-antenna vs multi-antenna systems, spatial and polarization diversity, channel capacity, MIMO performance metrics.

Antenna prototyping [3 lectures]
Materials used in antenna fabrication, substrate selection, traditional prototyping method, printable dielectric and conductive materials, 3D printing for fabrication, challenges in 3D printing.

Textbooks

1. Balanis C. A., (2016), *Antenna Theory, Analysis and Design*, 4thedition, Wiley Press.
2. Krauss J. D., (2006),*Antennas and Wave Propagation*, 4thedition, TMH Education.
3. Elliott R. S., (2006), *Antenna Theory and Design*, Wiley Press.
4. Volakis J. L., (2019), *Antenna Engineering Handbook*, 5thedition, McGraw Hill.
5. Chen Z. N., (2016), *Handbook on Antenna Technologies*, Springer.
6. Gao S., Luo Qi, Zhu F., (2014), Circularly Polarized Antennas, Wiley-IEEE Press.

Self-Learning Materials

1. Kumar, G., *Antennas*, NPTEL Course Material, Department of Electrical Engineering, Indian Institute of Technology Bombay, https://onlinecourses.nptel.ac.in/noc17_ee03/preview
2.Ferraro, A., *Antenna Engineering*, Video Lectures, Penn State Univ., USA, http://www.engr.psu.edu/cde/courses/ee438/index.htm.

Preparatory Course Materials

1.Kumar, P., *Electromagnetic Theory*, NPTEL Course Material, Department of Electrical Engineering, Indian Institute of Technology Kanpur, https://nptel.ac.in/courses/108104087/.

2. Kumar, G., *Microwave Theory and Techniques*, Department of Electrical Engineering, Indian Institute of Technology Bombay, https://nptel.ac.in/courses/108101112/

Title	RFIC and MMIC	Number	ECL7XX
Department	Electronics Engineering	L-T-P [C]	3–0–0 [3]
Offered for	B.Tech. (IV Year), M.Tech., and PhD	Type	E
Prerequisite	Analog electronics, Communication engineering, Microwave engineering		

Objectives

1. To introduce the students with the integrated circuit designing at radio frequencies.
2. Introduction of various circuits and building blocks for communication applications.

Learning Outcomes

1. Understanding the design of integrated circuits at high frequencies.
2. Practical implementation of various circuits required to build a transceiver system.
3. Learning about the design of various active components like mixers, LNAs, power amplifiers at RF and microwave frequencies.

Contents

Introduction to RFIC & MMIC technology [8 lectures**]**
Component Technology and Foundry Choice, passive and active Components, substrate material, wafer processing, CMOS, MESFET, HEMT, HBT modelling.
Foundry Use and Economics [4 lectures**]**
MMIC Production Costs. Simulation and Component Models-Passive and Active Component Models. Process Design for Product Yield. Sensitivity and Tolerance Analysis.
On chip Design [15 lectures**]**
Inductors, transformers, transmission line, varactors, constant capacitors. Active Baluns, Switch Design-Phase shifters. Switched-Path Attenuators, LNA topologies, gain switching, band switching, high IP3 LNA, Passive and active downconverter mixer, up converter mixer, self mixing LO**,** Design-Large Signal Effects on Active Devices, Power Amplifier, Power budget, Thermal management, SWaP-C (size, weight and power-cost)
MMIC Layout [12 lectures**]**
Layout Files, preparation, The Circuit Layout Process, Layout Checking, Chip Arraying, Processing Technology-Substrate Material Growth. Wafer Production. Surface Layers. Photolithography. Wafer Thinning. Through-Substrate Vias. Chip Separation. Quality Assurance. Test-Process Control & Monitoring. DC Test and Stability Problems. RF Test and Calibration**.**

Textbooks

1. T. H. Lee, the Design of CMOS Radio-Frequency Integrated Circuits. Cambridge, UK: Cambridge University Press, 1997.
2. B. Razavi, RF Microelectronics, 2nd Edition, Pearson Education, 2014.
3. Practical MMIC Design by Steve Marsh (Artech House)

Self Learning Materials

1. RF Integrated Circuits, NPTEL, Dr. S. Chatterjee, IIT Delhi, http://nptel.ac.in/courses/117102012/#
2. High Speed Communication Circuits, MIT, https://ocw.mit.edu/courses/electrical-engineering-and-computer-science/6-776-high-speed-communication-circuits-spring-2005/lecture-notes/

Preparatory Course Materials

1. B. P. Lathi, Z. Ding, Modern Digital and Analog Communications Systems, 4th edition, Oxford university press, July-2017.
2. D. M. Pozar, Microwave Engineering, 4th edition, New York, NY, USA: Wiley, 2011.

Title	Millimeter-wave and THz Technologies	Number	ECL7XX
Department	Electronics Engineering	L-T-P [C]	3–0–0 [3]
Offered for	B.Tech., M.Tech., and PhD	Type	E
Prerequisite	Microwave engineering		

Objectives
1. Introduction of the phenomenon and circuit characteristics at mm-wave frequency ranges.
2. introduce various guiding structures, antennas, and switches at mm-wave frequencies.

Learning Outcomes
1. Understanding the design problems and challenges at mm-wave frequencies.
2. Ability to design various mm-wave guiding structures, passive and active circuits.

Contents
Introduction to millimeter-wave and THz technology [5 lectures**]**
Mm-wave and THz frequency bands, Terahertz gap, atmospheric losses, Propagation and path losses, Friis equation, link budget and power budget calculation, applications in wireless communication, security, and space, quasi-optical system, properties of gaussian beam.
Guiding structures for mm-wave/THz circuits [10 lectures**]**
losses in planar circuits (conductor loss, leakage loss, dielectric loss, surface waves), Air filled waveguides, dielectric guides, NRD guide, substrate integrated waveguides (SIW), Substrate Integrated NRD (SINRD), dielectric image line, planar air filled waveguide, Waveguide and planar resonators, dispersion, circular dielectric waveguide, polymer/plastic waveguides, plasmonic waveguides, loaded, unloaded, and external Q-factors
Beam-forming network [8 lectures**]**
Hybrid coupler, crossover, phase shifter, butler matrix, Nolen matrix, Rotman lens, beamforming antennas, hybrid and digital beamforming techniques, adaptive phased arrays.
Advanced antennas [7 lectures**]**
Problem with traditional antennas, SIW based slot and cavity antenna, leaky wave antenna, waveguide slot array, on chip antennas, surface-wave antenna, metasurface antennas, nano-antennas, wearable antennas.
Terahertz sources and detectors [6 lectures**]**
Semiconductor-based sources, THz pulse generation using photo-mixing, free electron laser, diode-based detectors, photoconductive antennas, heterodyne and up-conversion detection.
Selected topics in THz [3 lectures**]**
Terahertz wireless communication, THz Channel Propagation Characteristics, Terahertz time-domain spectroscopy (THz-TDS), mm-wave/THz Sensing and NDE, Metamaterials and metasurfaces for THz applications.

Textbooks
1. Koul S. K., (1997), Millimeter Wave and Optical Dielectric Integrated Guides and Circuits, Wiley-Blackwell
2. Courtney D. Liu, B. Gaucher, U. Pfeiffer, and J. Grzyb, (2009), Advanced millimeter wave technologies, Wiley.
3. Rieh Jae-Sung, (2021), Introduction to Terahertz Electronics, Springer.

Self Learning Material
1. Mandal M. K., Millimeter-wave technology, NPTEL course material, Department of Electronics and Electrical Communication Engineering, IIT Kharagpur, https://nptel.ac.in/courses/117105139/
2. TI Training & Vedios, (2017), mmWave RADAR in ADAS Applications, https://training.ti.com/mmwave-radar-adas-applications

Preparatory Course Material

1. D. M. Pozar, (2011), Microwave Engineering, 4th Ed. New York, NY, USA: Wiley.

Title	RF System Design for Communications	Number	ECL7XX0
Department	Electronics Engineering	L-T-P [C]	3–0–0 [3]
Offered for	B.Tech., M.Tech., and PhD	Type	E
Prerequisite	Engineering Electromagnetics, Communication systems, RF Circuits		

Objectives

1. Familiarize behaviour of cascaded blocks in RF system and effect of specification of blocks on system performance.
2. Provide basic understanding about behaviour of intermodulation products, spurious responses.
3. Provide basic understanding of linearity of a system.

Learning Outcomes

1. Formulate and analyse system specification to control gain requirement, noise figure of the system.
2. Obtain basic understanding of linearity of a cascade from specifications on its modules.
3. Obtain basic understanding of the spurious responses in a conversion scheme.

Contents

Introduction to wireless communication systems [6 lectures**]**
Review of antenna concepts, Radio Channel and modulation requirements, bits, symbols and waveforms, wireless system building blocks, system specification, quadrature amplitude modulation (QAM), quadrature phase shift keying (QPSK)

Gain [5 lectures**]**
T and X parameter, Module Gain, Overall Response, Standard Cascade, Bilateral Modules, small and large signal conditions

Noise [4 lectures**]**
Fundamentals of noise, Noise Figure (NF), NF of cascaded systems, noise floor, impedance dependence of Noise Figure, phase noise, noise floor, sensitivity, SFDR

Nonlinearities [5 lectures**]**
Representing Nonlinear responses, gain compressions, harmonics and intermodulation (IM) products, measuring IMs, AM-PM and PM-AM conversions, non-linearities of cascaded stages

Transceiver architectures [14 lectures**]**
Receiver architecture, super heterodyne, direct down conversion and low IF receivers, problem of images, self-image, flicker noise, direct RF-sampling receiver, down sampling, transmitter architectures, heterodyne and direct conversion transmitters.

Contaminating Signals [5 lectures**]**
Frequency conversion basics, spurs, two-signal IMs, hard & soft limiting, mixing through the LO Port - AM Suppression, FM Transfer, single-sideband transfer, mixing between LO components.

Textbooks

1. William Egan, (2003), Practical RF System Design, by William Egan, 6th Edition, John Wiley & Sons
2. Ulrich L. Rohde, David P. Newkirk, (2000), RF/ Microwave Circuit design for wireless applications, John Wiley & Sons
3. B. Razavi, RF Microelectronics, 2nd Edition, Pearson Education, 2014.

Self Learning Material

1. Kumar P., RF Integrated Circuits, Department of EE, Indian Institute of Technology Madras, https://nptel.ac.in/courses/117102012/#

Title	RADAR Engineering	Number	ECL7XX0
Department	Electronics Engineering	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech., M.Tech., and PhD	Type	E
Prerequisite	Engineering Electromagnetics, Communication systems, RF Circuits		

Objectives
1. Teaching the fundamentals RADAR.
2. Teaching building blocks and different types of RADAR.

Learning Outcomes
The students are expected to have the ability to:
1. Understand the building blocks and working principles of RADAR.

Contents
Introduction [4 lectures**]**
Working principle, RADAR range equations, maximum unambiguous range, pulse repetition frequency, radar block diagram, Probability of detection, False alarm, radar cross section of targets.
Types of RADAR [10 lectures**]**
Pulse radars and CW radars, MTI RADAR, over the horizon RADAR, SAR and ISAR, Doppler RADAR, delay-line cancellers, blind speeds.
Moving target and tracking [8 lectures**]**
Moving target detector, limitations of MTI, tracking with radar, monopulse tracking, conical scan.
RADAR measurements [8 lectures**]**
RADAR measurement and accuracy, Range and velocity ambiguities, the ambiguity diagram, static and dynamic RCS, monostatic and bistatic measurements.
Pulse shaping [4 lectures**]**
Pulse compression-principles, matched filter, chirp waveforms, Descriptions of land & sea clutter, detection of targets in clutter.
RADAR transceiver [5 lectures**]**
Transmitter architecture, isolation between transmitter and receiver, duplexers, Receiver architecture, heterodyne receiver, noise figure.

Textbooks
1. Skolnik M.I., (2017), *Introduction to RADAR systems*, 3rd Ed., McGraw Hill.
2. Richards M.A., (2005), *Fundamentals of RADAR signal processing*, Indian Ed., McGraw Hill.
3. Carpentier M.H., (1988), *Principles of modern RADAR systems*, Artech house publishers.

Self Learning Materials
O'Donnell R.M. , *Introduction to RADAR systems*, MIT open course, MIT Lincoln Laboratory, https://ocw.mit.edu/resources/res-ll-001-introduction-to-radar-systems-spring-2007/

Title	Optical Signal Processing	Number	ECL7XX0
Department	Electronics Engineering	L-T-P [C]	3–0–0 [3]
Offered for	B.Tech., M.Tech., and PhD	Type	E
Prerequisite	Electromagnetism and optics, Signals and systems		

Objectives
1. Introduction to the basics of nonlinearities in optical devices and how they can be harnessed for processing of light waves.

2. Explaining the working, modelling and characterization of various optical signal processing devices along with their advantages and limitations.

Learning Outcomes
1. Ability to design, characterize and analyze basic optical signal processing circuits.
2. Appreciate the advantages of optical signal processing over conventional electronic signal processing techniques.

Contents
Nonlinear optics for signal processing [15 lectures**]**
Nonlinear optical susceptibility, time-domain description of optical nonlinearities, Kramers-Kronig relations, intensity dependent refractive index of materials
Third order nonlinearities [15 lectures**]**
Theory and modeling of Optical Kerr Effect, Self Phase Modulation and Cross Phase Modulation processes, Optical phase conjugation, Four Wave Mixing and Third Harmonic Generation, Photon-phonon interaction, Stimulated Brillouin, Raman and Rayleigh scattering, two Photon Absorption.
Optical signal processing devices [9 lectures**]**
Optical Fourier Transform, optical correlators, pulse compression, pulse shaping, optical interconnects, multiplexers, mixers, wavelength conversion, tunable optical delay.

Textbooks
1. Boyd, R. W., (2013), *Nonlinear optics*. Elsevier.
2. Dragoman D., and Dragoman M. (2013), *Advanced optoelectronic devices*. Vol. 1. Springer Science and Business Media..
3. Saleh, B.E.A, and Teich M.C., (1991), *Fundamentals of photonics*. Vol. 22. New York: Wiley.

Self-Learning Materials
1. Singh B.P and Rustagi K.C., *Nonlinear Optics,* NPTEL Course Material, Department of Physics, Indian Institute of Technology Bombay,
https://nptel.ac.in/courses/115/101/115101008/

Preparatory Course Materials
1. Roy S., *Introduction to Nonlinear Optics and its Applications,* NPTEL Course Material, Department of Physics, Indian Institute of Technology Kharagpur,
https://nptel.ac.in/courses/115/105/115105105/

Title	**Computational Electromagnetics**	**Number**	**ECL7XX0**
Department	Electronics Engineering	L-T-P[C]	3–0–0 [3]
Offered for	B. Tech., MTech., and PhD	Type	E
Prerequisite	Engineering Electromagnetics		

Objectives
1. Introduce the basic analytical and numerical techniques in electromagnetics.
2. familiarize the students with different numerical techniques used in commercial electromagnetic solvers.

Learning Outcomes
1. Solve the electromagnetic problems analytically and numerically.
2. To understand the commercial simulators like HFSS, CST, and COMSOL.

Contents
Review of Maxwell's equation [7 lectures**]**
Boundary condition, wave equation, separation of variables in different coordinate system, orthogonal functions, Strum-Liouville differential equation, Green's Function
Finite Difference methods [6 lectures**]**
Finite difference approximations, boundary conditions, finite differencing of different PDEs, accuracy and stability of FD methods, some practical examples.
Finite Difference Time Domain (FDTD) analysis [10 lectures**]**
Wave equation and solution for waveguide modes, Yee algorithm, numerical dispersion and stability, common source waveforms for FDTD simulations, FDTD analysis in 1D, FDTD in 2D/3D, Absorbing Boundary Conditions (ABC), Perfectly Matching Layer (PML), dispersive materials modelling, practical examples.
Finite Element Method (FEM) [10 lectures**]**
Discretization, governing equations, basis functions, FEM analysis in 1D, boundary conditions, accuracy and numerical dispersion, solution of 2D wave equation, mesh generation.
Method of Moments (MoM) [6 lectures**]**
Introduction to MoM, basis functions, classification of integral equations, solution of integral equations using MoM, thin wire approximation, thin wire excitations, Hallen integral equation, Pocklington equation.
Assignments to be given on MATLAB and commercial software.

Textbooks
1. Garg R., (2008), *Analytical and Computational Methods in Electromagnetics*, Artech House.
2. Sadiku M. N. O., (2018), *Computational Electromagnetics With MATLAB*, 4th Edition, Taylor and Francis Group.

Title	Photonic Devices and Circuits	Number	ECL7XX
Department	Electronics Engineering	L-T-P [C]	3–0–0 [3]
Offered for	B.Tech. (IV Year), M.Tech., and PhD	Type	E
Prerequisite	Engineering Electromagnetics		

Objectives
1. To introduce the students with various Lasers and Photodetectors, and integrated photonic circuit elements.

Learning Outcomes
1. Understanding of the structure, design and working of various photonic devices.
2. Knowledge about the theoretical model and practical implementation of the photonic devices and circuits.
3. Ability to design photonic integrated circuits using the basic building blocks .

Contents
Sources [11 lectures]
Light matter interaction, Spontaneous and stimulated emission, Einstein Coefficients, Line Broadening, Theory of laser amplification, Amplifier Pumping, Common laser amplifiers, nonlinearity, noise in amplifiers, laser oscillation, Types of lasers: Pulsed lasers, Q switching, Mode locking.
Rate equations and operating characteristics, transient response, modulation response, noise characteristics, external optical feedback effects in semiconductor lasers.
Photodetectors [10 lectures]

Materials, heterostructures, p-i-n, avalanche and Schottky junction photodiodes, gain and responsivity, single photon APDs, Array detectors, noise in photodetection: spectral density functions, shot noise, johnson noise, avalanche noise.

Integrated Photonic Circuits [10 lectures]

Materials and fabrication techniques for Photonic Integrated circuits, Propagation in rectangular dielectric waveguides: effective index method, losses. Elements of integrated circuits: Waveguide Couplers, Y branches, Angular and Circular bends, Crossovers, Polarization control, multimode interference devices, Diffraction Gratings, Wavelength filters and modulators.

Machine Learning for Photonics [8 lectures]

Introduction to Machine Learning in Photonics; Regression Analysis, Accuracy Assessment; Introduction to Neural Networks: Lost Function, Forward & Backward Propagation; Inverse Photonic Design with NNs.

Textbooks

1. B.E.A. Saleh, M.C. Teich, Fundamentals of Photonics, Wiley.
2. Amnon Yariv, Pochi Yeh, Photonics: Optical Electronics in Modern Communication, Oxford University Press.
3. Richard Osgood Jr., Xiang Meng, Principles of Photonic Integrated Circuits, Springer.

Title	**Optical Communication Systems**	Number	**ECL7XX0**
Department	Electronics Engineering	L-T-P[C]	3–0–0 [3]
Offered for	B.Tech., M.Tech., and PhD	Type	E
Prerequisite	Communication systems		

Objectives

1. Provide students an understanding of the concepts related to transmission and reception. techniques for high speed optical communication systems.

Learning Outcomes

1. Analyze basic and high speed optical communications systems.
2. Compare and contrast the features of various optical communication techniques.

Contents

Introduction to optical communication systems [2 lectures]

Basic building blocks, optical transmitters, optical receivers, optical channel, optical amplifiers.

Optical channel considerations [8 lectures]

Attenuation, dispersion, pulse broadening and chirping, line coding schemes, eye pattern, channel noise effects, turbulence models, weather conditions viz. fog, haze, drizzle, under-water impairments.

Lasers and modulation [12 lectures]

Coherence, population inversion and feedback, spontaneous and stimulated emission, injection LASERS, gain and index guided LASERS, DFB, DBR Lasers. LiNbO3 - MZ, III-V semiconductor EAM modulators, high speed silicon modulators: operating principles, design, performance criterion, modulator drivers, reliability and bias control.

Photo-detection [5 Lectures]

P-I-N, APD, responsivity, quantum efficiency, noise, SNR, NEP, bandwidth, low, high and transimpedance receivers.

System performance [10 Lectures]

Principles of Coherent and Incoherent systems, intensity modulation/direct detection system, SIM, homodyne and heterodyne detection, SNR, BER performance in coherent and non-coherent modulation schemes and channel impairments, Rise time and Power budgeting, channel capacity.

Textbooks

1. Keiser, G. (2017), *Optical Fiber Communications*, 5th Edition, McGraw Hill.
2. J. M. Senior, Optical Fiber Communications: Principles and Practice, Prentice Hall, 3rd Edition, 2010.
3. Chadha, D. (2013), *Terrestrial Wireless Optical Communications*, 1st Edition, McGraw Hill.
4. Chen A. and Murphy, E. (2011), *Broadband Optical Modulators: Science, Technology and Applications*, 1st Edition, CRC Press.
5. Jia-Ming Liu (2016), Principles of Photonics, 1st Edition, Cambridge University Press.

Self Learning Materials

1. Kumar, K. P., *Fiber-Optic Communication Systems and Techniques*, NPTEL Course Material, Department of Electrical Engineering, Indian Institute of Technology Kanpur, https://nptel.ac.in/courses/108/104/108104113/

Title	Optical Networks	Number	ECL7XX0
Department	Electronics Engineering	L-T-P [C]	3–0–0 [3]
Offered for	B.Tech., M.Tech., and Ph.D	Type	E
Prerequisite	Basics of Fiber Optic Communications		

Objectives

1. Introduction to the elements and components of an optical networking solution.
2. Exploring the capabilities and limitations of different network topologies.

Learning Outcomes

1. Ability to identify the required network topology for a given requirement.
2. Solving capability of WDM network design problems.

Contents

Introduction to Optical Networks [8 lectures**]**
Types of traffic, Services, optical packet and circuit switching, optical layered architecture. Optical Network elements

Local Area Network [6 lectures**]**
Network topologies, multiplexing and multiple access techniques, network resource sharing, network capacity, Medium Access Protocols.

Optical Access Networks [8 lectures**]**
PON architecture, EPON, GPON, WDM-PON, multi point control protocol, dynamic bandwidth allocation, FTTx networks

Metropolitan and wide area networks [6 lectures**]**
SONET/SDH, Optical Transport Network, Wavelength Routing Network, Routing and Assignment Algorithm.

Network control, management and survivability [6 lectures**]**
Network management system architecture, Functionality, GMPLS, Impairment, Survivability, Protection and Restoration.

Free-Space optical networks [5 lectures**]**
Classification, FSO network topologies, Design factors, FSO Network applications.

Textbooks

1. Chadha D. (2019), *Optical WDM networks: from static to elastic networks*, Wiley-IEEE Press.
2. Grobe K. and Eiselt M. (2013), *Wavelength Division Multiplexing, A Practical Engineering Guide*, Wiley.
3. Ramaswami R., Sivarajan K.N. and Sasaki G.H. (2010), *Optical Networks: A Practical Perspective*, 3rd Edition, Morgan Kaufmann Publishers.

Preparatory Course Material

1. Keiser G. (2013), *Fibre Optic Communication*, McGraw-Hill, 5th Edition.

Title	IR Detectors and Imaging	Number	ECL7XX0
Department	Electronics Engineering	L-T-P [C]	3–0–0 [3]
Offered for	B.Tech., M.Tech., and Ph.D	Type	E
Prerequisite	Basics of Electromagnetism and Optics		

Objectives
The instructor will:
1. Introduce the fundamentals of IR detection and various types of detectors.
2. Introduce the processing of information in the form of images through IR elements.

Learning Outcomes
The students are expected to have the ability to:
1. Use various detectors for their design of an IR system
2. Bridge the gap between theory and practice with real time image processing using IR elements.

Contents
IR Basics [10 lectures**]**
Infrared radiation, irradiance scattering, spectral bands, materials, coatings, blackbody radiation, impact of environmental conditions.
IR Detectors [14 lectures**]**
Characteristics, performance parameters: noise, responsivity, sensitivity. thermal detectors: thermopile, bolometers, pyroelectric detectors. photon detectors: photoconductive, photovoltaic, PN, PIN, APD, Schottky-Barrier diode, metal-semiconductor-metal, photo emissive detectors, III-V detectors, HgCdTe Detectors, IV-VI detectors, quantum well and superlattice detectors, focal plane array architectures.
Imaging [15 lectures**]**
Fourier transformation, spatial frequency spectrum, Fraunhofer and Fresnel diffraction with various apertures, Huygen's principle, Kirchoff's Integral, Rayleigh criterion, Fresnel-Kirchhoff Theory. Concept of coherence, mutual coherence function, propagation of mutual coherence function. fourier transform property of lens, image formation, coherent information processing systems: spatial filtering, area modulation, optical correlation, applications in image restoration, SAR, LIDAR and processing of broadband signals.

Textbooks
1. Antonio Rogalski , Infrared Detectors, CRC Press.
2. Infrared Thermal Imaging, Wiley, 2018 Michael Vollmer Klaus-Peter Möllmann.
3. Franscis T.S. Yu, Optical Information Processing, Wiley Interscience, 1983.
4. Max Born and Emil Wolf, Principles of Optics, Cambridge University Press.

NAC/Feedback

Comments by the experts

NAC members			
Name of the Expert	**Organization**	**Comments**	**Actions taken**
Prof. B K Kanaujia,	NIT Jalandhar	The designed course work is good and covers all major topics.	**1.** S-parameters and signal flow graphs are now

NAC Chairman		Furthermore, some points can be considered. **1.** Basic of RF and Microwave (S-parameters, Signal Flow graphs) seem to be missing or very brief. These concepts are very important to understand any RF design and should be included in the 1st semester. **2.** The existing Microwave Circuit Design syllabus can be further enriched by adding a dedicated unit on Directional Couplers and Power Dividers. These components are fundamental building blocks in microwave engineering and are widely used in RF front-ends, measurement setups, and communication systems. **3.** The current syllabus of Advanced Electromagnetics largely revisits the fundamental concepts of electromagnetic theory. To make the course more advanced and aligned with current research and industry needs, additional modern computational techniques may be included such as Finite-Difference Time-Domain (FDTD) method. **4.** There is a notable overlap of transmission line concepts between the Computational Electromagnetics (CEM) course and the Advanced Electromagnetics course. **5.** Electives on niche areas such as Microwave Imaging, RF Energy Harvesting, Smart Antenna scan be included."	included in "Microwave Circuits" **2.** Couplers and power dividers are now included under the "Microwave Circuits" course. **3.** All these topics are included in the computational electromagnetic course. **4.** Transmission lines are excluded from the "Advanced Electromagnetics" course. **5.** Thank you for the suggestions. However, due to the limitations on the number of elective courses, these courses are not included.
Prof. Satish Sharma	San Diego State University, USA	**1.** Why so many textbooks? Generally, it is good to have ONE textbook and the rest as recommended books. How can you ask students to buy so many books? **2.** All courses should include the number of lectures for each chapter so a better teaching plan can be made. **3.** Some courses propose content that may not be possible to cover in a semester. I have identified these in the document.	**1.** As per your suggestion, a single text book will be suggested in the class. **2.** The suggestion has been incorporated. **3.** Some topics can be omitted during the lectures and the same can be given to students for self-study. **4.** Punctuation has been checked throughout the document. **5.** Included.

		4. English punctuation is needed throughout the document. **5.** Optical waveguide in fiber optics **6.** Microwave circuits – start with S parameters **7.** Radio wave propagation – on top before dipole **8.** Remove S parameters from RF system design **9.** Noise before non-linearity – RF system design **10.** Add antenna component in RF system design **11.** Wafer we can include in RFIC and MMIC **12.** Layout can be changed to MMIC layout in RFIC and MMIC course **13.** Beam forming can be omitted in mm-wave technology **14.** Advanced topics – filters and antennas.	**6.** S-parameters are now included **7.** Radio wave propagation is now included before dipole antenna. **8.** S-parameters are now removed from RF system design. **9.** Noise is now included before non-linearity. **10.** Review of antenna design is added in the RF system design course. **11.** Wafer processing is now included in the RFIC and MMIC course. **12.** Changed as suggested. **13.** Beam forming has not been omitted as it is suggested by one industry expert. **14.** Filter design is one dedicated course in our curriculum and advanced antennas are covered in mm-wave and THz technologies course.
Dr. Abhishek Awasthi	Paras Anti Drone	**1.** Industrial Engagement (in terms of summer internship) **2.** Introduction of Mixed Signal RF Circuits (Basic Idea about high frequency ADC/DAC Chips and their application). **3.** Basic Introduction of Adaptive phased array (Beamforming and beam tracking)	**1.** M.Tech projects will be floated with some industrial collaborations. **2.** These topics are covered under "RFIC and MMIC" design and "RF System Design" courses. **3.** Covered under "millimeter-wave and terahertz technologies" course.
Dr. Prashant Vashishth	Defence Lab Jodhpur	**1.** The curriculum appears to be very comprehensive, covering all of the fundamentals and advanced topics in microwave and photonics, and is well aligned with the basic fundamentals and covering futuristic scope in these areas. **2.** Some topics need to be included in RADAR Engineering: MHz/Over the horizon radar, SAR and ISAR imaging algorithms, Static and dynamic radar cross section measurement	**1.** Thanks for the appreciation. **2.** The suggested topics are now included in the syllabus of RADAR Engineering. **3.** Suggested topics are now included in the syllabus of "Microwave and optical measurement". 4. WIll be discussed in the classes along with regular lectures. **5.** Thank you for your kind support.

		techniques, mono-static and bi-static measurement **3.** Contemporary techniques like coherent radar measurement from drone based radars, ultra narrow pulse measurement in the course "Microwave and optical measurement". **4.** Broad potential applications should be mentioned in each course. **5.** The course appears to be in line with DLJ Jodhpur's ongoing and futuristic research area, and students can do internships with DLJ as part of their thesis work. Further any support required for M tech aspirants in terms of collaborative research wok and training will be imparted at DLJ.	
Prof. S. K. Varshney	IIT Kharagpur	**1.** You may rename: Microwave Filters & Filtering Networks as Microwave Filters & Networks **2.** Semiconductor lasers as an independent elective might not be suitable as it can be part of either optical fiber communication or optical communication systems. **3.** You may add Optical Sensors as an alternative to this, as an elective **4.** Silicon Photonics: More topics can be introduced into this, bring Micro-ring resonators, and on-chip routing, silicon photonics for neuromorphic computing etc **5.** Instead of Optical fiber Communication, I'd rather rename it as: Fiber Optic Components and Devices as there is another subject with Optical Communication Systems. Under this Fiber Optic Components and Devices, amplifier technologies (EDFA, SOA, Raman) can be brought in.	**1.** The subject name has been renamed as suggested. Finally this course is dropped due to limitations in the number of electives. **2.** Semiconductor Laser Course is removed as an independent elective. The main concepts are distributed in Optical Communication Systems and Photonic devices and Circuits courses. An additional elective titled Quantum Optics and Engineering (from already approved MTQT course) is included to cover the concepts in quantum domain. **3.** NA. Electives may be added at later stages based on need. **4.** Additional topics are included as suggested. **5.** The Course on Optical Fiber Communication is renamed to Fiber Optic Devices and Components. Many devices and components such as FBGs, Isolators, Circulators, MMI devices, mode field convertors, Wavelength MUX/DEMUX, fiber to chip

			coupling, Fiber Amplifiers: EDFA, Raman and Brillouin, SOA are included.
Prof. Amol Choudhary	IIT Delhi	**1.** The name of the program can be looked into Microwave and Optical is a bit outdated in my opinion. Can the name itself be aligned in some way to the national missions or thrusts? Perhaps focusing on integration or systems? **2.** The program structure itself needs to be looked into. 32 credits out of 56 are for the thesis which is more of an MSR structure focussed on research. For training a skilled pipeline for industry and upcoming missions, I think you need to introduce internships, lab courses I couldn't see any lab course. The measurement course and microwave circuits course did have some lab component. But at least 6 credits should be for labs). Thesis should ideally be not more than 20 credits. **3.** Can you introduce a design course? Students have to design a product or instrument from scratch. They would also have to include commercials. etc. This would be very helpful from an industrial point of view. **4.** Generally, the course content seems to be a bit traditional. Can all the content be looked into to make it more 2026? For example, in optical. communications, DSP, Coherent, SDM may be introduced. 15 lectures for lasers can be reduced to 10-12.	**1.** As per the suggestion, the name of the program is changed to microwave and photonic technologies. **2.** The program structure is already approved and cannot be changed. The comment regarding the internship is noted and will be explored. Credit structure is already approved. **3.** Core courses are already approved and cannot be changed. Also, due to the limitation on the number of elective courses, the design course cannot be included. However, to introduce some design components, the design assignments will be given in microwave and optical measurement course. **4.** DSP is already included in optical signal processing course.

Experts from Industries

Name of the Expert	Organization	Comments	Actions taken
Dr. Kranti Kumar Katare	Ericsson Antenna	Thank you for reaching me out. I just went through the attached curriculum; it is looking quite	As per the following suggestion, following action has been taken:

	Technology, Germany	relevant to the industry and on-going research work. The suggested topics from my side which are in high demand in industries (can also be included for the curriculum, If possible (not mandatory) **1.** Artificial intelligence/Machine learning **2.** MIMO **3.** 3D printed antennas **4.** Antenna for biomedical applications (nano antennas, wearable antennas) **5.** Microwave/THz sensors	Added MIMO antennas in Antennas and Wave propagation, Nano-antennas, wearable antennas in mm-wave and THz technologies under advanced antennas, Terahertz sensing was already included – this is changed to mm-wave/THz sensing. Some introductory part of AI/ML was already included in photonic devices and circuits. Topic on 3D printed antenna is now included in Antennas and Wave Propagation course under antenna prototyping.
Ms. Srimoyi	Analog Devices, USA	The curriculum looks good overall and matches industry expectations. Below is the synopsis of the additions that may be done in the curriculum: Beamforming is widely used in various communication and defense applications. For example, its used in 5G cellular networks where massive MIMO antennas are used to focus data beams towards cellphones, improving data rates. Another example could be a scenario where beamforming could be used in drones. When there is a jammer, it will intercept with the drones receiving capability. So, beamforming technology is used in drones and direct high intensity beams towards the required direction thus overriding the jammers capability to intercept. As per the curriculum, phase shifter, hybrid coupler and other analog beamforming techniques are introduced. Would be good to add a few digital and Hybrid beamforming techniques as well in this course. Would be good to add topics related to advanced modulation schemes used in LTE and 5G. Other thing is on practical aspects of system design (may be a few practical examples on how blocker and jammers can interfere	As per the suggestion, following action has been taken: Quadrature amplitude modulation (QAM), quadrature phase shift keying (QPSK) in RF system design under introduction to wireless communication systems, SWaP-C (size, weight and power-cost) in RFIC and MMIC under on-chip design, Hybrid and digital beamforming techniques in mm-wave and Thz technologies under beam forming networks. We have added an elective course on Electromagnetic Interference and Compatibility. Component selection, PCB layout and its challenges will be covered in EMI/EMC course.

		with signal reception and techniques to mitigate those), EMI/EMC challenges (couldn't find this in the curriculum), Component selection, PCB Layout and its challenges, thermal and RF shielding techniques, SWaP-C (Size, Weight, and Power-Cost). I understand that most of these topics are covered in the course but would be good if these practical scenarios are added as well while teaching.	
Dr. Vinay Narayane	Paras Anti-drone	The course contents are aligned with industry needs.	Thank you for your response.

Feedback from Education Policy Committee (EPC) Meeting

Comments	Action taken
The Committee suggested to add B.E./B.Tech. in Electrical engineering or equivalent degrees with valid GATE score in EC (for non-CFTI candidates) as eligibility criteria along with other already mentioned branches. For CFTI candidates, GATE score is not mandatory for CGPA 8/10 or above.	The eligibility criteria have been revised accordingly.

Feedback from the Senators

Comments	Action taken
The eligibility criteria for candidates with fellowship and without fellowship should be clearly defined and specified.	Revised and clearly mentioned for different categories of admissions.
The Department should consider including Infra-Red applications	A separate elective course "**IR Detectors and Imaging"** has been added in the revised curriculum.

SPGC meeting 19.05.2026

Comments	Action taken
Remove M.Sc. candidates under the admission category with MoE fellowship for direct offering with CGPA above 8.0 without GATE	Revised.

Conclusion and recommendation:

This proposed M. Tech. program in Microwave and Photonic Technologies is designed to meet the growing needs of important fields like telecommunications, defence, aerospace, and quantum technologies. The program combines strong fundamental concepts with advanced research and hands-on design experiences to get future engineers ready to come up with new ideas and excel in designing high-frequency and high-speed circuits and systems critical to national security and next generation communication systems.